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Practical 1

1.Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

Execution task:

```
jamuna@Jem:~$ nano practical1.c
jamuna@Jem:~$ gcc practical1.c
jamuna@Jem:~$ ./a.out
Enter a command:
ls

Parent PID:923
Child PID:932

Child Process completed.

Child PID:932
Parent PID:815
a      ac         dir5           execl1.c    f1.c  fork.c         linux_task4    permission_lab   practical1.c
a.c    b1         dir6           execlp.c    f2    fork2.c        newfile1       pipe.c           read1.c
a.out  b2         execcv.c       execlp1.c   f3    fork2.c.save   open2.c        pipe2.c          readwrite.c
a7     content.c  execcv.c.save  execv.c     f4    fork3.c        open2.c.save   pipex.c          student_portal
a8     d          execl.c        f1          fi    fork3.c.save   pen2.c         pipex2.c         tt
```

```
jamuna@Jem:~$ uname
Linux
```

```
jamuna@Jem:~$ lscpu
Architecture:             x86_64
  CPU op-mode(s):         32-bit, 64-bit
  Address sizes:          39 bits physical, 48 bits virtual
  Byte Order:             Little Endian
CPU(s):                   12
  On-line CPU(s) list:    0-11
Vendor ID:                GenuineIntel
  Model name:             13th Gen Intel(R) Core(TM) i5-1334U
    CPU family:           6
    Model:                186
    Thread(s) per core:   2
    Core(s) per socket:   6
    Socket(s):            1
    Stepping:             3
    BogoMIPS:             4992.00
    Flags:                fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr ss
                          e sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology tsc_reliable
                           nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid sse4_1 sse4_2 x2api
                          c movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowpre
                          fetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi
                          1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv
                          1 xsaves avx_vnni vnmi umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_cl
                          ear serialize ibt flush_l1d arch_capabilities
Virtualization features:
  Virtualization:         VT-x
  Hypervisor vendor:      Microsoft
  Virtualization type:    full
Caches (sum of all):
  L1d:                    288 KiB (6 instances)
  L1i:                    192 KiB (6 instances)
  L2:                     7.5 MiB (6 instances)
  L3:                     12 MiB (1 instance)
NUMA:
  NUMA node(s):           1
  NUMA node0 CPU(s):      0-11
Vulnerabilities:
  Gather data sampling:   Not affected
  Ghostwrite:             Not affected
  Indirect target selection: Not affected
  Itlb multihit:          Not affected
  L1tf:                   Not affected
  Mds:                    Not affected
  Meltdown:               Not affected
  Mmio stale data:        Not affected
  Old microcode:          Not affected
  Reg file data sampling: Mitigation; Clear Register File
  Retbleed:               Mitigation; Enhanced IBRS
  Spec rstack overflow:   Not affected
  Spec store bypass:      Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:             Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:             Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSB-eIBRS SW sequence; BHI BHI_D
                          IS_S
  Srbds:                  Not affected
  Tsa:                    Not affected
  Tsx async abort:        Not affected
  Vmscape:                Not affected
```

```
jamuna@Jem:~$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda    8:0    0 356.9M  1 disk
sdb    8:16   0 159.4M  1 disk
sdc    8:32   0     2G  0 disk [SWAP]
sdd    8:48   0     1T  0 disk /mnt/wslg/distro
                               /
```

```
jamuna@Jem:~$ ps
    PID TTY          TIME CMD
    816 pts/0    00:00:00 bash
    997 pts/0    00:00:00 ps
```

```
jamuna@Jem:~$ top
top - 03:45:43 up 19 min,  1 user,  load average: 0.05, 0.04, 0.00
Tasks:  24 total,   1 running,  23 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7762.6 total,   7126.9 free,    545.9 used,    240.9 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   7216.7 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0   24140  15388  11604 S   0.0   0.2   0:00.97 systemd
      2 root      20   0    3180   2212   2072 S   0.0   0.0   0:00.02 init-systemd(Ub
      6 root      20   0    3180   2044   1940 S   0.0   0.0   0:00.00 init
     49 root      19  -1   50400  16784  15504 S   0.0   0.2   0:00.22 systemd-journal
     81 systemd+  20   0   22416  14484  11984 S   0.0   0.2   0:00.08 systemd-resolve
     87 root      20   0   35364  12328   9240 S   0.0   0.2   0:01.17 systemd-udevd
    107 root      20   0    2888   1948   1820 S   0.0   0.0   0:00.07 chronyd-starter
    108 root      20   0    4472   3140   2892 S   0.0   0.0   0:00.01 cron
    109 message+  20   0    8820   5452   4772 S   0.0   0.1   0:00.08 dbus-daemon
    115 root      20   0   44096  29892  14752 S   0.0   0.4   0:00.31 networkd-dispat
    150 root      20   0   18436   9412   8168 S   0.0   0.1   0:00.09 systemd-logind
    177 syslog    20   0  220548   5936   4660 S   0.0   0.1   0:00.16 rsyslogd
    218 root      20   0    5492   2776   2588 S   0.0   0.0   0:00.01 agetty
    232 _chrony   20   0   23684  11196   7224 S   0.0   0.1   0:00.11 chronyd
    240 _chrony   20   0   12096   2436   1796 S   0.0   0.0   0:00.00 chronyd
    273 root      20   0  123460  32464  17856 S   0.0   0.4   0:00.18 unattended-upgr
    334 root      20   0    8648   5584   4740 S   0.0   0.1   0:00.02 login
    379 jamuna    20   0   22336  13512  10888 S   0.0   0.2   0:00.24 systemd
    381 jamuna    20   0   22912   4088   2088 S   0.0   0.1   0:00.00 (sd-pam)
    411 jamuna    20   0    6136   5436   3864 S   0.0   0.1   0:00.07 bash
    814 root      20   0    3184   1124    980 S   0.0   0.0   0:00.00 SessionLeader
    815 root      20   0    3200   1260   1108 S   0.0   0.0   0:00.04 Relay(816)
    816 jamuna    20   0    6268   5600   3928 S   0.0   0.1   0:00.13 bash
   1007 jamuna    20   0    8180   6016   3900 R   0.0   0.1   0:00.00 top
```